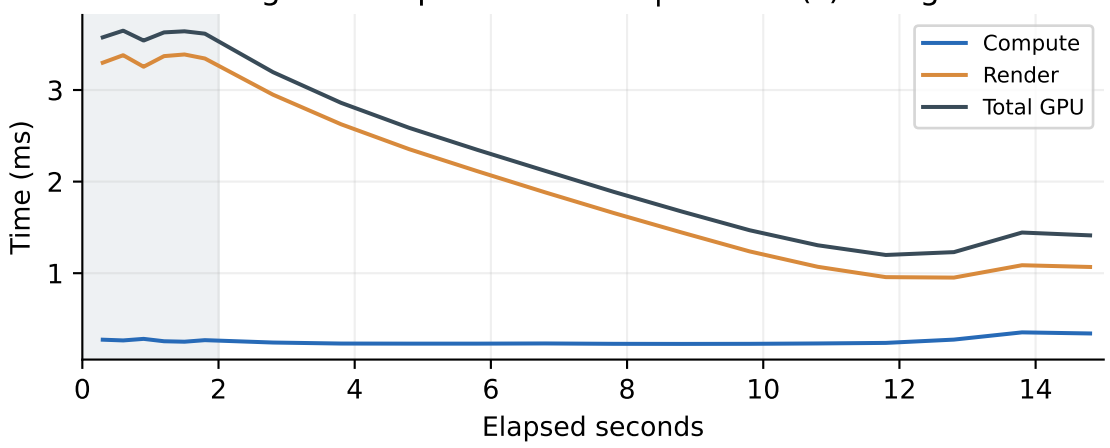
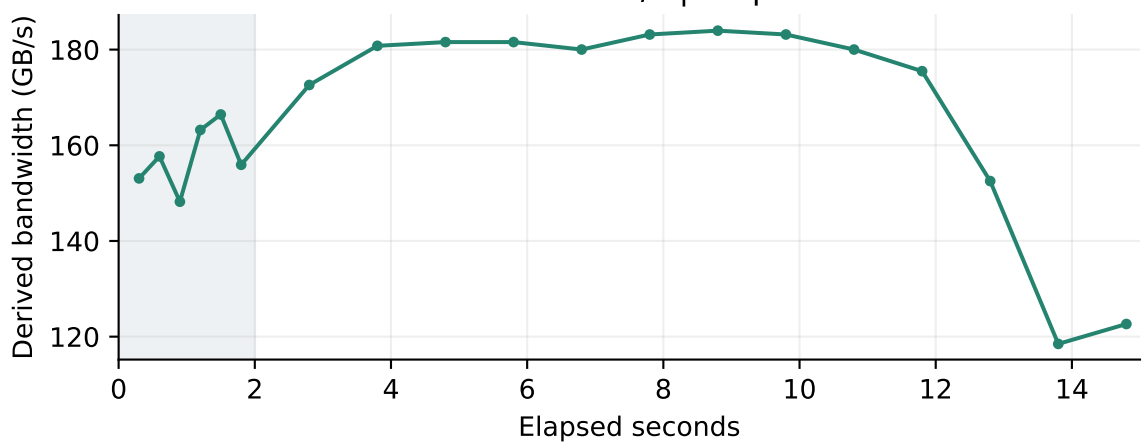
M4 Pro Vulkan/MoltenVK control runs | 1M particles | 15 seconds

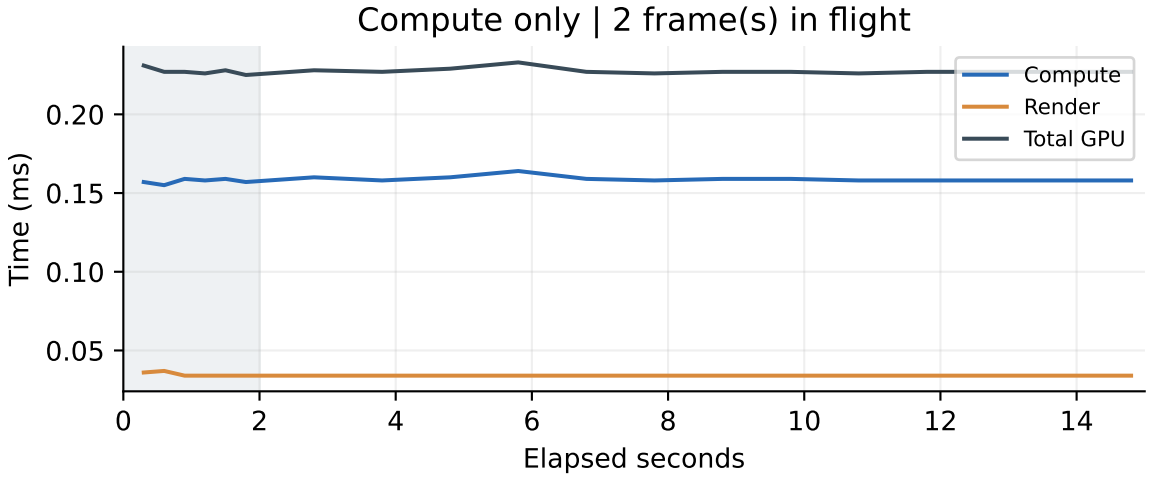
Original compute + render | 1 frame(s) in flight



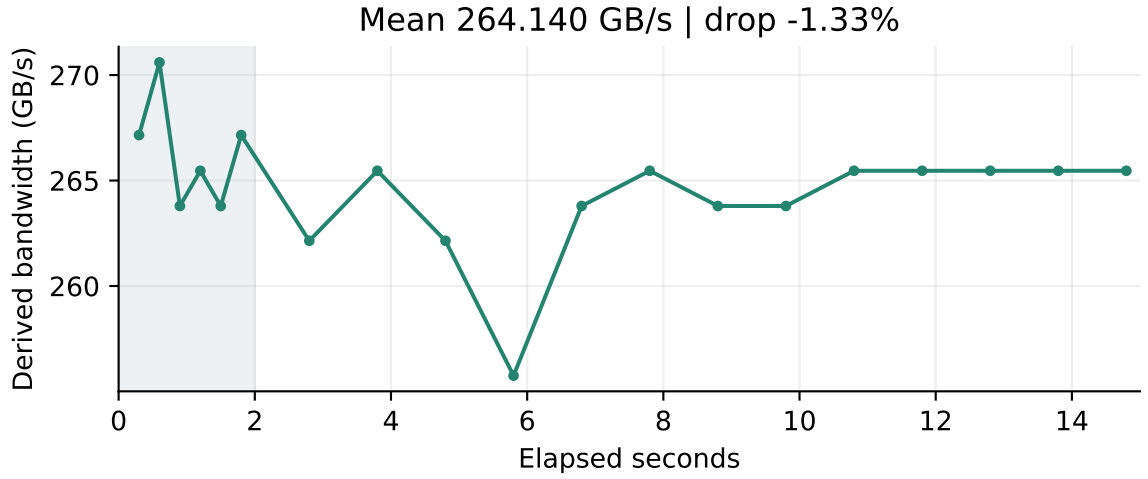
Mean 164.016 GB/s | drop 16.54%



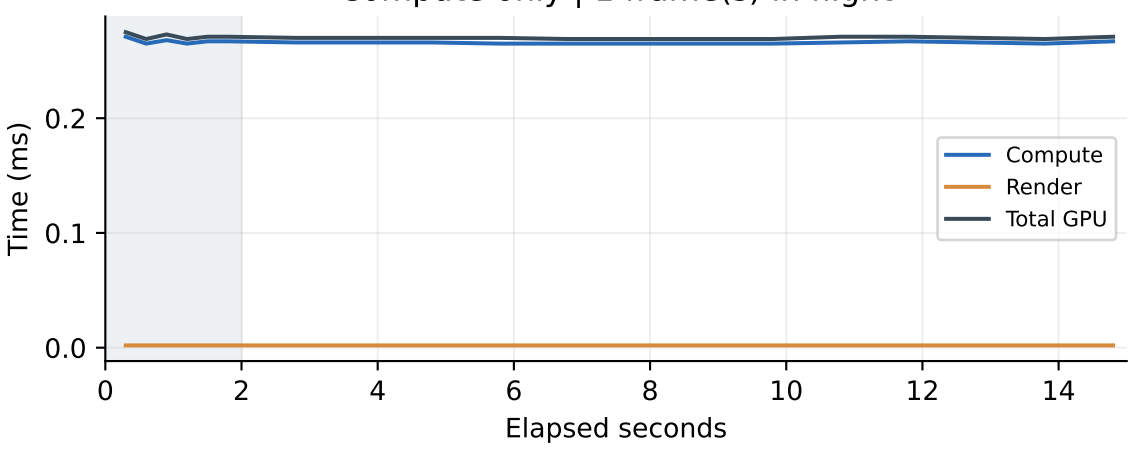
Compute only | 2 frame(s) in flight



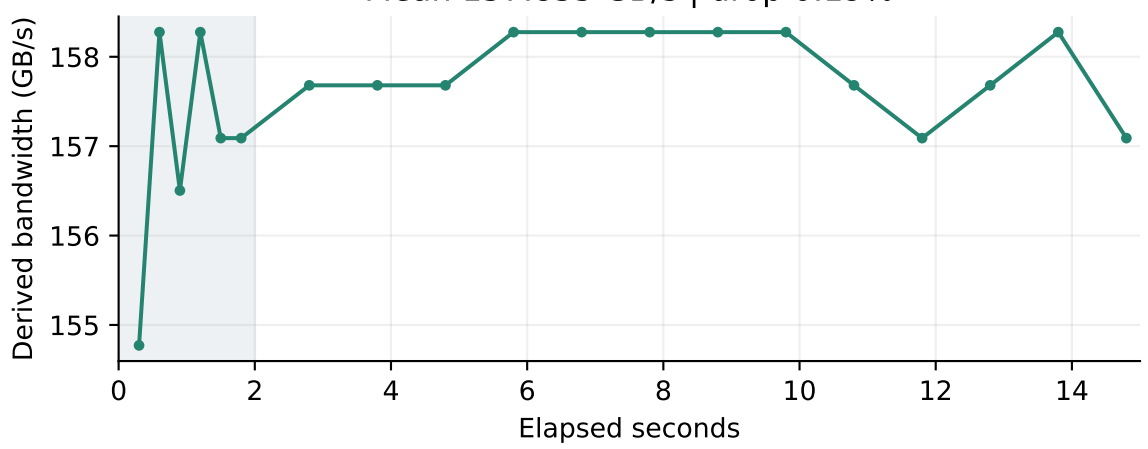
Mean 264.140 GB/s | drop -1.33%



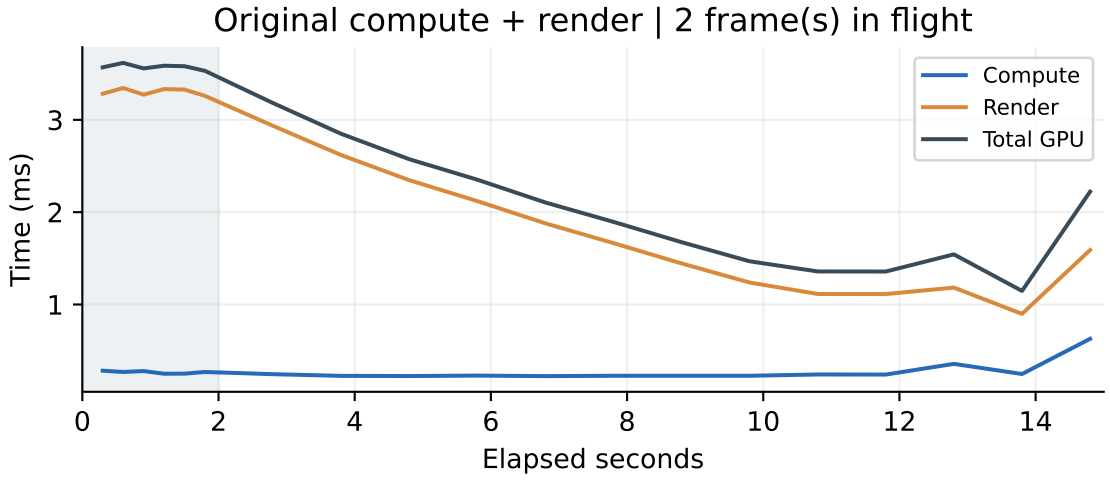
Compute only | 1 frame(s) in flight



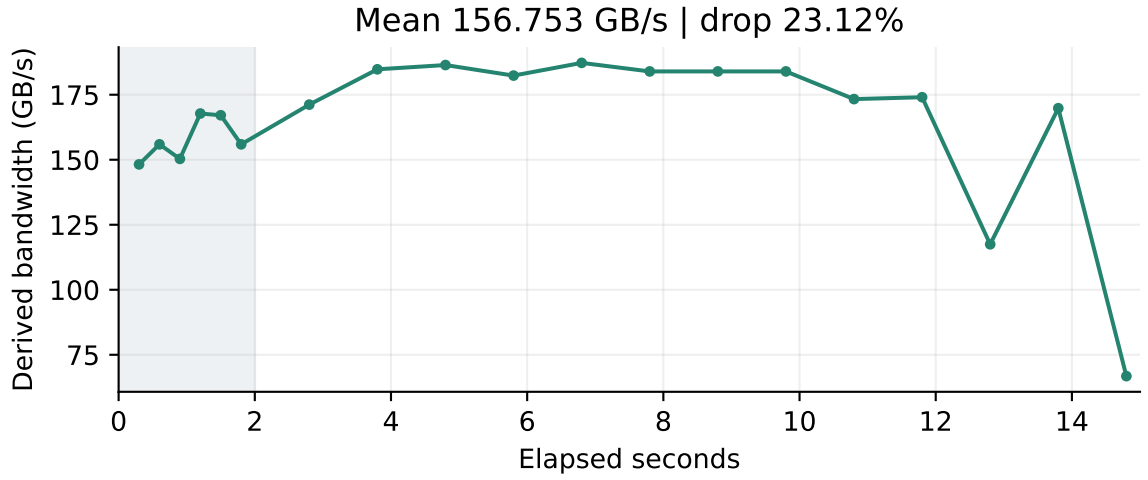
Mean 157.835 GB/s | drop 0.19%



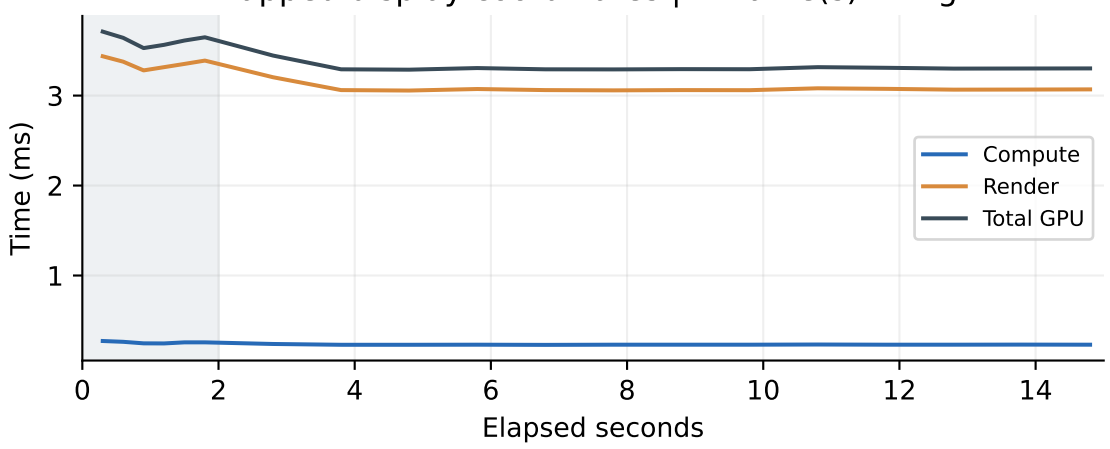
Original compute + render | 2 frame(s) in flight



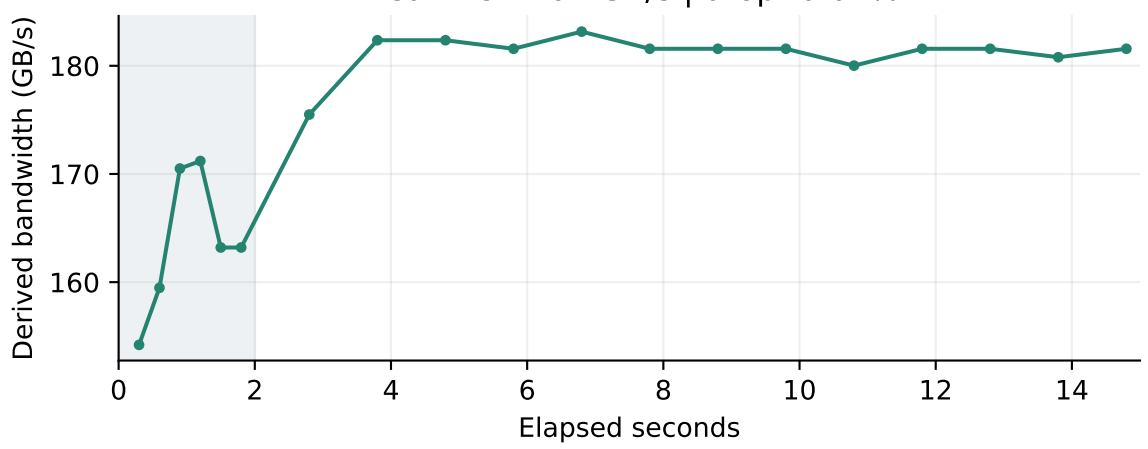
Mean 156.753 GB/s | drop 23.12%



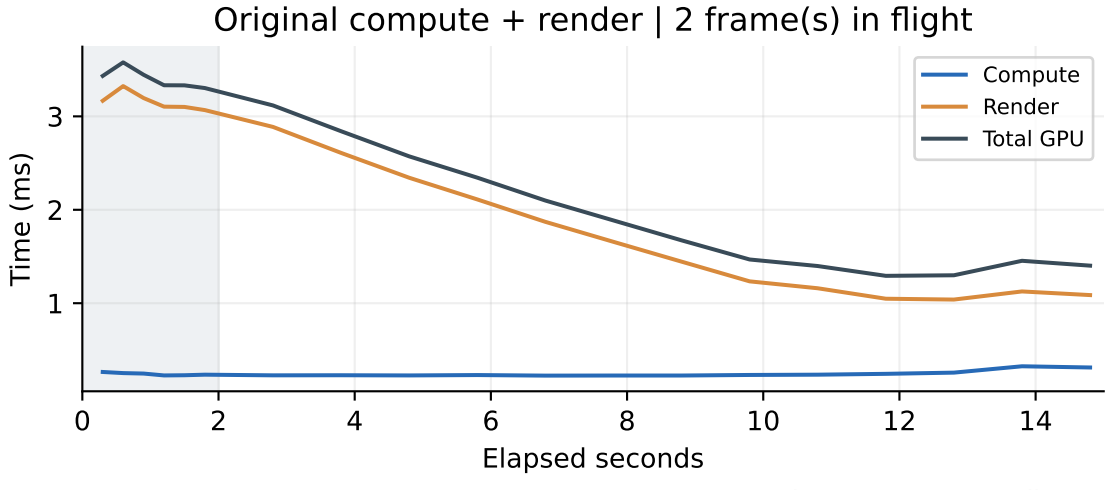
Wrapped display coordinates | 2 frame(s) in flight



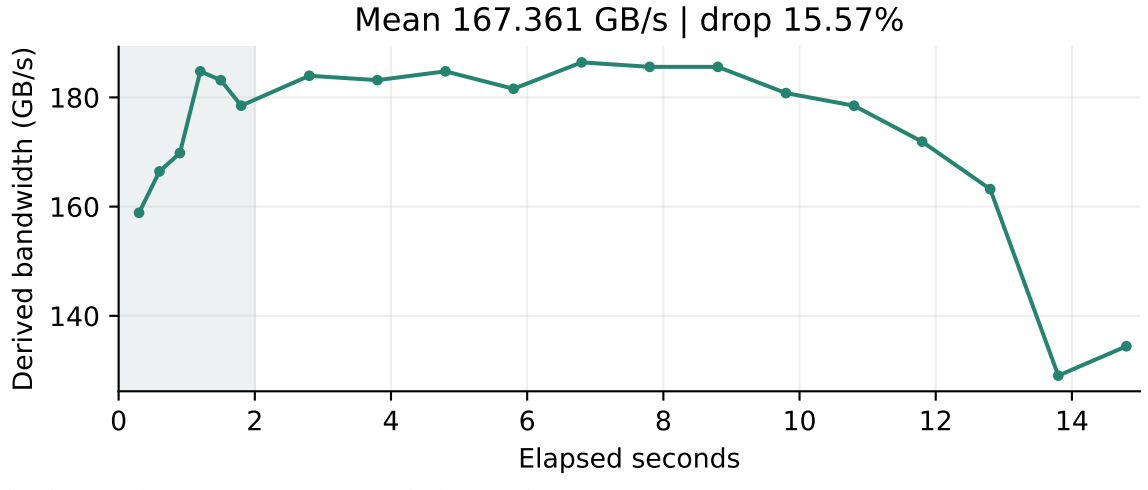
Mean 181.467 GB/s | drop -0.01%



Original compute + render | 2 frame(s) in flight



Mean 167.361 GB/s | drop 15.57%



Grey: 2s warm-up. Variants are separate diagnostic observations. Curves use rounded log values.